

Matrix functions and stability

# Decidability of strict stability for rational matrix families

**Difficulty:** extreme **Importance:** broadly interesting**Status:** Open

**Rating rationale:** Extreme because termination at an unpromised strict threshold is a foundational decidability barrier; broad impact includes switched control and algorithmic matrix-product theory.

## Context and notation

The joint spectral radius of a nonempty compact set  $\mathcal{M} \subset \mathbb{C}^{d \times d}$  is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written  $\rho(A)$ .

## Problem statement

Does a Turing machine exist which, on every input consisting of a finite nonempty list of matrices in  $\mathbb{Q}^{d \times d}$ , encoded by binary integer numerators and positive denominators, halts and correctly decides whether

$$\hat{\rho}(\mathcal{M}) < 1?$$

Dimension and list length are part of the input. No separation from the threshold is promised. The question asks for termination and correctness, without a polynomial running-time requirement.

## References and status evidence

Jungers, *The Joint Spectral Radius: Theory and Applications*, §2.2.3, Open Question 1, printed p. 29 of the author manuscript. Blondel and Tsitsiklis, *The boundedness of all products of a pair of matrices is undecidable*, Systems & Control Letters 41 (2000), 135–140, §2, proves undecidability of the non-strict test  $\hat{\rho} \leq 1$ . That theorem does not answer the strict test.

The screen included “joint spectral radius” with “strict”, “decidability”, “less than one”, and 2025/2026. No deciding algorithm or applicable undecidability reduction was located. Some later literature summarizes the non-strict theorem as a generic stability impossibility; admission here follows its actual threshold. MF-04 is a proposed finite-product property, whereas this entry asks for a decision procedure even if such a property fails.

## Audit — 2026-09-10

Rechecked the indexed text of Jungers, Open Question 1 and strict-stability/decidability searches. No deciding algorithm or applicable undecidability theorem was located. Daubechies–Lagarias’ correction independently distinguishes the unresolved subunit test from the undecidable unit-threshold problem. Evidence remains historical and bounded.